

Pascal Corso

Dr sc. ETH Zurich & Mechanical Engineer

*Specialised in Machine Learning & Scientific Computing,
Data Analysis & Biomedical Engineering,
Computational & Experimental Fluid Dynamics*



Belgian-Italian | Based in Zürich, Switzerland | Born 31 July 1990 | [Professional portfolio](#)
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PROFESSIONAL SUMMARY

Researcher, mechanical engineer and entrepreneur with experience in machine learning, high-performance computing (HPC), data analysis, biomedical engineering and computational and experimental fluid mechanics.

Key competencies:

- Advanced **numerical simulations** (CFD, FSI, FEM) and mathematical modelling
- **Scientific computing and machine learning** (Python, MATLAB, C++, FORTRAN)
- Design and analysis of **laboratory flow experiments**
- **Biomedical flow data analysis and medical imaging**
- Development and **optimisation** of cardiac **valve prostheses**
- **Management** of **multidisciplinary** research and development (R&D) **projects**

Career highlights:

- 10+ years' experience in computational & experimental fluid mechanics, biomedical engineering, data analysis, scientific and HPC.
- 4+ years' experience in data science & machine learning.
- 10+ articles published in peer-reviewed leading journals about aortic flows, turbulence, heart valves, medical imaging.
- 15+ international conference presentations spanning fluid mechanics, turbulence, biomechanics, computer methods.
- Leader on projects about cardiovascular devices, fluid & solid mechanics and scientific computing.

My expertise in computational fluid dynamics, biomedical data analysis and machine learning enables me to tackle complex engineering challenges. Having successfully supported or led **10+ multidisciplinary research projects** while **mentoring 10+ students and researchers**, I excel at developing and translating technical innovations into practical applications for biomedical and clinical contexts.

LEADERSHIP

Head of the computational haemodynamics sub-group

Jan. 2021 – Dec. 2023

University of Bern – ARTORG Center for Biomedical Engineering Research

As head of the computational haemodynamics research sub-group concurrent with the Postdoctoral Researcher role below, I provided scientific mentorship to three PhD students, defining their research topics and establishing strategic research priorities for the group. I coordinated regular meetings to foster collaboration among team members and external scientists, organised scientific activities including seminars and training sessions to support continuous professional development and group growth and created an environment conducive to high-quality research output within the computational haemodynamics field.

WORK EXPERIENCE

Hi-D Imaging AG

Sept. 2026 – Feb. 2027

Experimental Flow Research Engineer (50%, concurrent with independent research)

- Performing two-dimensional Particle Image Velocimetry (2D-PIV) and three-dimensional Particle Tracking Velocimetry (3D-PTV) flow characterisation on collaborator cardiovascular devices.

- Contributing to ISO 5840/17025-aligned test protocol development.
- Supervising interns and junior engineers on experimental methods.

ETH Zurich, Swiss Data Science Center

Oct. 2024 – Feb. 2027

Scientific Collaborator & Research Affiliate

In collaboration with two biology laboratories at École Polytechnique Fédérale de Lausanne (EPFL):

- Investigated spatial distributions of brain tissue lipids by using deep generative models applied to mass spectrometry imaging data, developing conditional variational autoencoders to enhance spatial resolution ([Link to GitHub repository](#)). My focus has since shifted to deep generative super-resolution of blood flow fields governed by physics constraints.

Alongside this, as independent research endeavours:

- Leading a line of research and scientific topics aiming to develop an *in silico-in vitro* optimisation platform for next-generation prosthetic aortic valves.

University of Bern

Nov. 2020 – Jan. 2024

Postdoctoral Researcher

I advanced computational biomechanics research programmes in cardiovascular fluid dynamics, specialising in heart valve haemodynamics and fluid-structure interactions.

In this role,

- Spearheaded the Fluid-Structure Interaction (FSI) studies examining how bioprosthetic aortic valve design influences haemodynamics and leaflet kinematics.
- Developed comprehensive analyses investigating haemodynamic transition-to-turbulence in various aortic valve types.
- Led the computational analysis of 3D-printed aortic valve blood and leaflet dynamics.
- Designed and implemented new computational frameworks for predicting valve calcification.
- Wrote and submitted research grants as Principal Investigator.
- Established new collaborations across disciplines.
- Provided scientific supervision for 5 master's and doctoral students.
- Disseminated research findings through peer-reviewed publications and presentations at international conferences.

I contributed my expertise to strategic collaborative projects:

- Computational investigation of complex-shaped snow particle aerodynamics: drag prediction and wake characteristics.
- Accelerated FSI simulation of blood flow and mitral heart valve biomechanics.
- Three-dimensional analysis of haemodynamic wall parameters in the carotid bifurcation using computer-augmented four-dimensional flow-sensitised Magnetic Resonance Imaging (4D Flow MRI).

University of Bern

May 2020 – Aug. 2020

Scientific Assistant

- Conducted a computational study on the fluid-structure interaction problem concerning bioprosthetic aortic valves, emphasising leaflet motion tracking and exploring the interaction between leaflet kinematics and flow dynamics in the sinuses and ascending aorta.

ETH Zurich

Oct. 2015 – Jan. 2020

Scientific Assistant & Doctoral Candidate

- Investigated the haemodynamic impact of aortic valve stenosis and prosthetic aortic valves, characterising disturbed flow patterns in the ascending aorta.
- Conducted high-fidelity direct numerical and large-eddy simulations to analyse laminar-turbulent transition in cardiovascular flows using NEK5000 and OpenFOAM computational frameworks.
- Implemented three-dimensional Particle Tracking Velocimetry (3D-PTV) to characterise complex haemodynamic structures *in vitro*, utilising anatomically-accurate aortic phantoms to investigate flow patterns downstream of transcatheter and stenotic aortic valves.

- Validated computational and experimental findings using time-resolved and flow-sensitised magnetic resonance imaging (4D Flow MRI), advancing methodologies for MRI-based quantification of turbulent kinetic energy losses and wall shear stress distributions.
- Provided scientific supervision for 4 master's research projects, ensuring methodological rigour and research excellence.
- Developed collaborative research networks in computational fluid dynamics, focusing on multi-scale flow simulation applications.
- Provided teaching support for numerical hydraulics course.
- Published my work in peer-reviewed journals and delivered presentations at international scientific conferences.

University of Mons

Jan. 2014 – Dec. 2014

Scientific Assistant

- Generated meshes within diverse artery geometries for computational simulations.
- Conducted numerical simulations to analyse aneurysmal blood flow dynamics, with and without the insertion of Flow-Diverter stent.
- Developed a particle tracking strategy to track the motion of particles, including red blood cells and thrombus within blood flow.
- Employed *in vitro* two-dimensional Particle Image Velocimetry (2D-PIV) technique to conduct measurements in phantom models of arteries, enabling the validation of numerical results.
- Provided supervision and guidance to 3 master's students in the completion of their theses.

VESUVIUS Group SA

July 2012 – Aug. 2012

Research Intern

- Developed an experimental protocol to accurately measure the thermal conductivity and diffusivity of the refractory lining within a casting ladle.

EDUCATION

Doctor of Science in Biomedical Flows and Engineering, ETH Zurich Switzerland, **Oct. 2020**

Doctoral dissertation: Computational and experimental investigation of disturbed aortic flows – novel methods to assess flow abnormalities due to severe stenoses and prosthetic aortic valves.

[Link to dissertation](#)

Master of Science in Mechanical Engineering, University of Mons, Faculty of Engineering Belgium, **June 2013**

Focus track: energy production systems

Master's thesis completed **summa cum laude**: Study of the flow in an intracranial aneurysm: Comparative analysis of PIV measurements and COMSOL Multiphysics simulations with and without Flow-Diverter stents.

Final year project (in collaboration with the artist Mr Jacques Rémus): Design and study of the mechanical and hydraulic parts of a musical fountain. **Caterpillar Inc. awarded a prize for creativity in the project.**

List of completed courses **magna cum laude**: Applied fluid mechanics, applied thermodynamics, combustion, compressible and incompressible aerodynamics, computational fluid dynamics, computational heat transfer, continuous and discrete time process control, machine design, mechanical manufacturing processes, statistics, properties of metallic alloys, thermal engineering, turbomachinery and engines.

Bachelor of Engineering Science, University of Mons, Faculty of Engineering Belgium, **June 2011**

List of completed courses **cum laude**: Chemistry, computer science and programming, dynamics of mechanical systems and vibrations, electricity and electronics, fluid mechanics, heat transfer, material science, mathematics, numerical analysis, physics, probability, technical drawing and metrology, structural mechanics, theoretical mechanics, thermodynamics.

Bachelor of Music, Higher Education School of Arts in Mons

Belgium, **Sept. 2014**

Focus track: piano performance

SELECTED PUBLICATIONS & CONFERENCES

Peer-reviewed journal articles (complete list on [Google Scholar](#))

Tagliavini G., Holzner M., **Corso P.**, (2025) Modal analysis reveals imprint of snowflake shape on wake flow structures. *International Journal of Multiphase Flow* 193:105365. [DOI](#)

Corso P., Obrist D. (2024) On the role of aortic valve architecture for physiological hemodynamics and valve replacement. Part I: flow configuration and vortex dynamics. *Computers in Biology and Medicine* 176:108526. [DOI](#)

Corso P., Obrist D. (2024) On the role of aortic valve architecture for physiological hemodynamics and valve replacement. Part II: spectral analysis and anisotropy. *Computers in Biology and Medicine* 176:108552. [DOI](#)

Tsolaki E., **Corso P. (co-first author)**, Zboray R. et al. (2023) Multiscale multimodal investigation and simulation of structural alterations in failed bioprosthetic heart valves. *Acta Biomaterialia* 169:138–154. [DOI](#)

Corso P., Walheim J., Dillinger H. et al. (2021) Toward an accurate estimation of wall shear stress from 4D flow magnetic resonance downstream of a severe stenosis. *Magnetic Resonance in Medicine* 86(3):1531–1543. [DOI](#)

Corso P., Giannakopoulos G., Gülan U., Frouzakis C.E., Holzner M. (2021) A novel estimation approach of pressure gradient and haemodynamic stresses as indicators of pathological aortic flow using subvoxel modelling. *IEEE Transactions on Biomedical Engineering* 68(3):980–991. [DOI](#)

Corso P., Gülan U., Cohrs N. et al. (2019) Comprehensive *in vitro* study of the flow past two transcatheter aortic valves: comparison with a severe stenotic case. *Annals of Biomedical Engineering* 47(9):2241–2257. [DOI](#)

Preprint under review

Corso P., Tagliavini G., Nestola M.G.C., Coulter F.B. (2026) Predicting calcification risk in prosthetic aortic valves: a hybrid physics-based and machine learning approach. Zenodo preprint, in peer review at *npj Digital Medicine*. [DOI](#)

Selected conference presentations and invited talks (5 of 15+)

17th World Congress on Computational Mechanics & 10th ECCOMAS, Munich, July 2026. [*Strain-based finite-time Lyapunov exponents for calcification risk classification in bioprosthetic and elastomeric aortic valves*](#) (minisymposium MS120A). Image entry [*Zone of bloody shear*](#) shortlisted as a finalist in the Arts & Science Contest.

Invited seminar at EPFL, Lausanne, March 2026 (host: Prof. S. Deparis). [*Prosthetic aortic valve performance assessment: FSI simulations, spectral analysis and FTLE-based calcification prediction*](#).

20th International Symposium on Computer Methods in Biomechanics and Biomedical Engineering, Barcelona, 2025. [*Deep generative super-resolution for enhanced mouse brain imaging and optical blood flow measurements*](#).

2nd European Fluid Dynamics Conference, Dublin, 2025. [*Modal analysis reveals aortic valve design effects on blood flow patterns*](#).

30th Congress of the European Society of Biomechanics, Zurich, 2025. [*Computational analysis of wall shear stress and strain-based metrics as calcification indicators in tissue and polymeric prosthetic aortic valves*](#).

Editorial activity

Guest Editor (2026) Special Issue “Bioinspired Valve Engineering and Cardiovascular Modeling”, *Biomimetics* (MDPI; CiteScore 6.2, IF 4.2). Submission deadline 31 October 2026. [*Special Issue*](#)

Peer review activity

Reviewer (2019–2026) for *Advanced Functional Materials*, *Annals of Biomedical Engineering*, *Artificial Organs*, *Computer Methods and Programs in Biomedicine*, *Computers & Fluids*, *Journal of Biomechanics*, *Physics of Fluids* and *Scientific Reports* (20+ reviews across 8 journals).

CONTINUING EDUCATION

Certificate of Advanced Studies in Entrepreneurial Leadership in Technology Ventures , ETH Zurich	Switzerland, Sept. 2026
One-day Course on Intellectual Property , University of Bern	Switzerland, Oct. 2024
Hands-on Machine Learning for Fluid Dynamics , von Karman Institute for Fluid Dynamics	Belgium, Feb. 2023
Bern Winter School on Machine Learning , University of Bern	Switzerland, Feb. 2022
FOMICS Summer School on Meshfree and Immersed Boundary Methods for Fluid-Structure Interaction , Università della Svizzera italiana (USI)	Switzerland, Sept. 2017
CISM-JMBC Joint Advanced School on Complex Flows and Complex Fluids , CISM - International Centre for Mechanical Sciences	Italy, May 2017
EXCITE Summer School on Biomedical Imaging , ETH Zurich	Switzerland, Sept. 2016
EuHIT Summer School on Turbulence , European High-Performance Infrastructures in Turbulence	Poland, July 2016

LANGUAGES

French (native), **Italian** (native), **English** (C2), **Dutch** (B2) and **German** (B1).

SKILLS & EXPERTISE

Operating systems

Linux Ubuntu

Microsoft Windows

Apple macOS

Office & typesetting tools

Microsoft Word, Excel, PowerPoint

LaTeX

Programming

MATLAB/Octave

C++

MPI

Python

FORTRAN

OpenMP

Development tools &

environments

JupyterLab

Vim

Docker

Sublime Text

Git

Conda

Computer-Aided Design (CAD) &

Finite-Element Modelling (FEM)

SOLIDWORKS

MOOSE

Abaqus/Standard-CAE

Mesh generation

SnappyHexMesh

Cubit/TRELIS

Computational Fluid Dynamics

(CFD) solvers

OpenFOAM

NEK5000

ANSYS Fluent

COMSOL Multiphysics

IMPACT & AV-FLOW (in-house)

Machine learning

PyTorch

TensorFlow

scikit-learn

Data analysis & visualisation

ParaView